



VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

Digital Systems Design

(BECE102L)

Fall Semester 2025-26

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Digital Systems Design–BECE102L

Rubrics

Assessment	Weightage
Case Study-1	10 Marks (Deadline tentatively before 15.10.25)
Quiz – 1	10 Marks
Quiz – 2	10 Marks
Continuous Assessment Test(CAT) – 1	15 Marks
Continuous Assessment Test(CAT) – 2	15 Marks
Final Assessment Test(FAT)	40 Marks
Total	100 Marks

Course Objectives

1. Provide an understanding of Boolean algebra and logic functions.
2. Develop the knowledge of combinational and sequential logic circuit design.
3. Design and model the data path circuits for digital systems.
4. Establish a strong understanding of programmable logic.
5. Enable the student to design and model the logic circuits using Verilog HDL.

Course Outcomes

1. Optimize the logic functions using and Boolean principles and K-map.
2. Model the Combinational and Sequential logic circuits using Verilog HDL.
3. Design the various combinational logic circuits and data path circuits.
4. Analyze and apply the design aspects of sequential logic circuits.
5. Analyze and apply the design aspects of Finite state machines.
6. Examine the basic architectures of programmable logic devices.

Syllabus

Module:1	Digital Logic	8 hours
• Boolean Algebra: Basic definitions, Axiomatic definition of Boolean Algebra, Basic Theorems and Properties of Boolean Algebra, Boolean Functions, Canonical and Standard Forms, • Simplification of Boolean functions. Gate-Level Minimization: The Map Method (K-map up to 4 variable), Product of Sums and Sum of Products Simplification, NAND and NOR Implementation. • Logic Families: Digital Logic Gates, TTL and CMOS logic families.		
Module:2	Verilog HDL	5 hours
• Lexical Conventions, Ports and Modules, Operators, Dataflow Modelling, Gate Level Modelling, Behavioral Modeling, Test Bench.		
Module:3	Design of Combinational Logic Circuits	8 hours
• Design Procedure, Half Adder, Full Adder, Half Subtractor, Full Subtractor, Decoders, Encoders, Multiplexers, De-multiplexers, Parity generator and checker, Applications of Decoder, Multiplexer and De-multiplexer. Modeling of Combinational logic circuits using Verilog HDL.		

Module:4	Design of data path circuits	6 hours
N-bit Parallel Adder/Subtractor, Carry Look Ahead Adder, Unsigned Array Multiplier, Booth Multiplier, 4-Bit Magnitude comparator. Modeling of data path circuits using Verilog HDL		
Module:5	Design of Sequential Logic Circuits	8 hours
- Latches, Flip-Flops - SR, D, JK & T, Buffer Registers, Shift Registers - SISO, SIPO, PISO, PIPO, - Design of synchronous sequential circuits: state table and state diagrams, Design of counters: Modulo-n, Johnson, Ring, Up/Down, Asynchronous counter. Modeling of sequential logic circuits using Verilog HDL.		
Module:6	Design of FSM	4 hours
Finite state Machine(FSM):Mealy FSM and Moore FSM , Design Example : Sequence detection, Modeling of FSM using Verilog HDL.		
Module:7	Programmable Logic Devices	4 hours
Types of Programmable Logic Devices: PLA, PAL, CPLD, FPGA Generic Architecture.		

➤ Text Book(s)

- ❑ M. Morris Mano and Michael D. Ciletti, Digital Design: With an Introduction to the Verilog HDL and System Verilog, 2018, 6th Edition, Pearson Pvt. Ltd..

➤ Reference Books

- ❑ Ming-Bo Lin, Digital Systems Design and Practice: Using Verilog HDL and FPGAs, 2015, 2nd Edition, Create Space Independent Publishing Platform.
- ❑ Samir Palnitkar, Verilog HDL: A Guide to Digital Design and Synthesis, 2009, 2nd edition, Prentice Hall of India Pvt. Ltd.
- ❑ Stephen Brown and Zvonko Vranesic, Fundamentals of Digital Logic with Verilog Design, 2013, 3rd Edition, McGraw-Hill Higher Education.

❖ Mode of Evaluation:

- Continuous Assessment Test, Digital Assignment, Quiz, Final Assessment Test

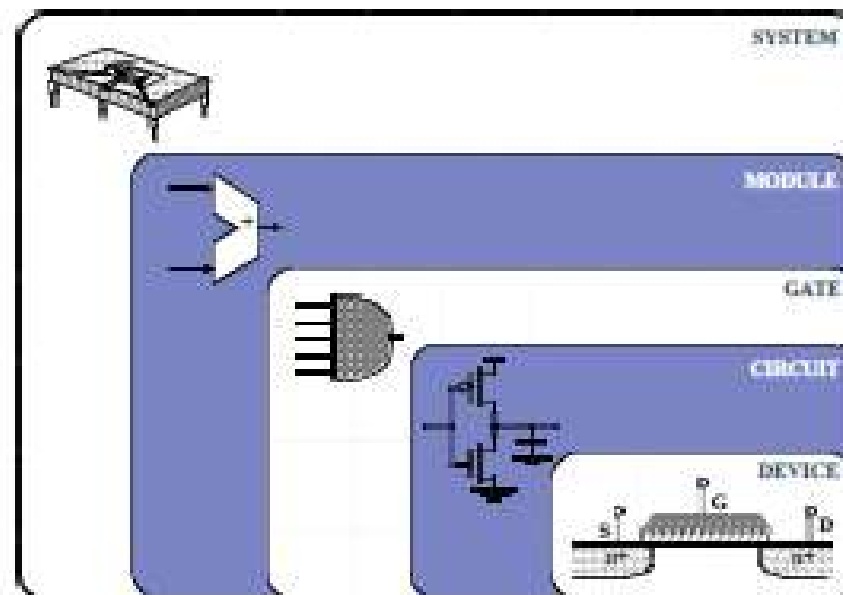
Module:1_Digital Logic

Discussion

- Boolean Algebra: Basic definitions, Axiomatic definition of Boolean Algebra, Basic Theorems and Properties of Boolean Algebra, Boolean Functions, Canonical and Standard Forms, Simplification of Boolean functions.
- Gate-Level Minimization: The Map Method (K-map up to 4 variable), Product of Sums and Sum of Products Simplification, NAND and NOR Implementation.
- Logic Families: Digital Logic Gates, TTL and CMOS logic families.

Digital Systems

- Digital systems have a prominent role in everyday life.
- Digital systems are used in communication, business transactions, traffic control, spacecraft guidance, medical treatment, weather monitoring, the Internet, and many other commercial, industrial, and scientific enterprises.
- **A digital system is a system that manipulates discrete elements of information represented internally in binary form.**
- A digital system is an interconnection of digital modules.



Logic Gates

- **Logic gates** are electronic circuits that operate on one or more input signals to produce an output signal.
- *Gate inputs and output* are driven/provides by voltages having two nominal values, e.g. 0 V and 5 V representing logic 0 and logic 1.

➤ Types of Logic gates

1) NOT/INVERTOR

2) AND

3) OR

4) BUFFER

5) XOR, XNOR

6) NAND

7) NOR

Basic logic gates

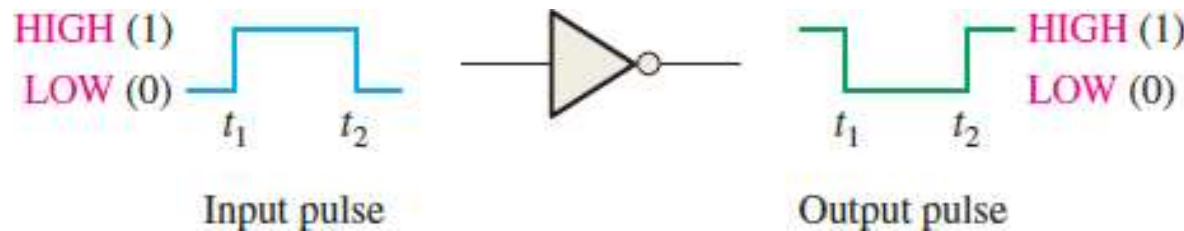
Universal logic gates (gates from which any digital circuit can be built)

The Inverter/ NOT gate

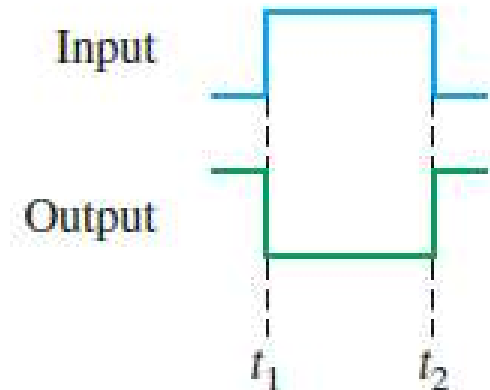
➤ Truth Table

Input	Output
LOW (0)	HIGH (1)
HIGH (1)	LOW (0)

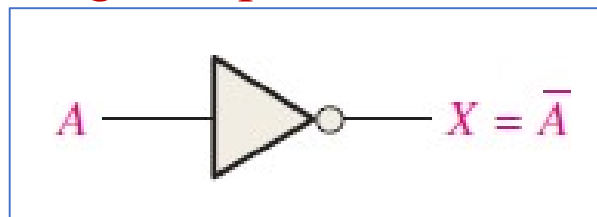
- When the input is LOW, the output is HIGH;
- When the input is HIGH, the output is LOW,
- Thereby producing an inverted output pulse.



➤ **Timing diagram** is basically a graph that accurately displays the relationship of two or more waveforms with respect to each other on a time basis.

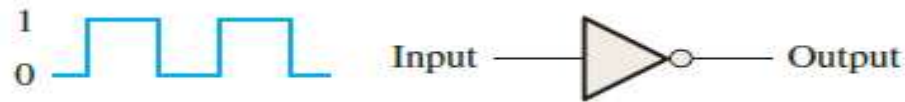


➤ Logic Expression for an Inverter:



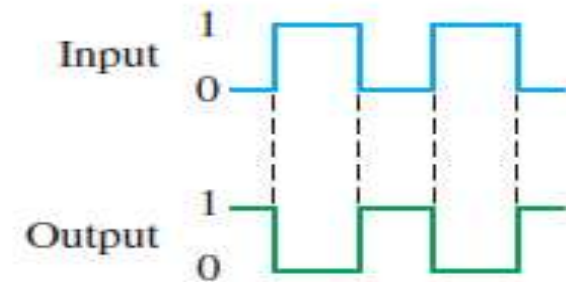
- The complement of a variable is designated by a bar over the letter.

- Example: A waveform is applied to an inverter as shown in Figure below.



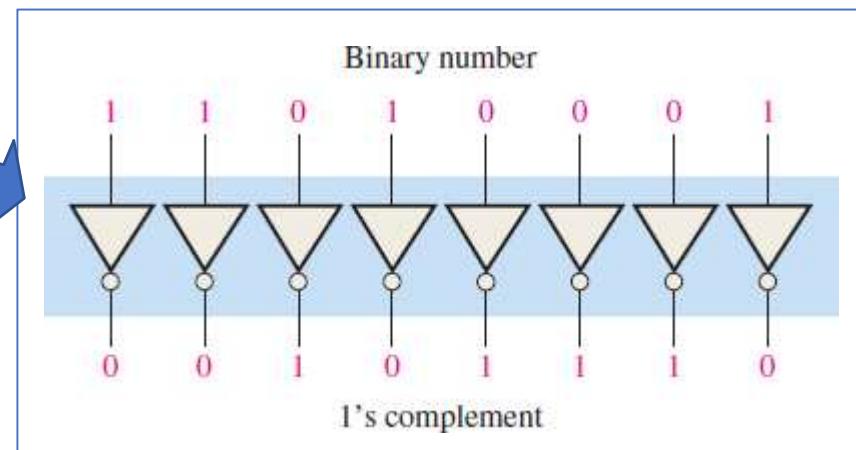
Determine the output waveform corresponding to the input and show the timing diagram. According to the placement of the bubble, what is the active output state?

- The output waveform is exactly opposite to the input (inverted), which is the basic timing diagram.
- The active or asserted output state is 0.



Application

Circuit for producing the 1's complement of an 8-bit binary number



AND Gate

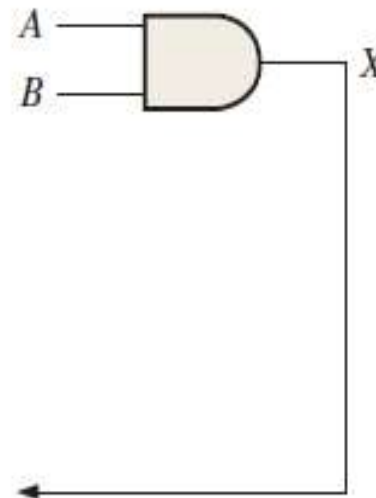
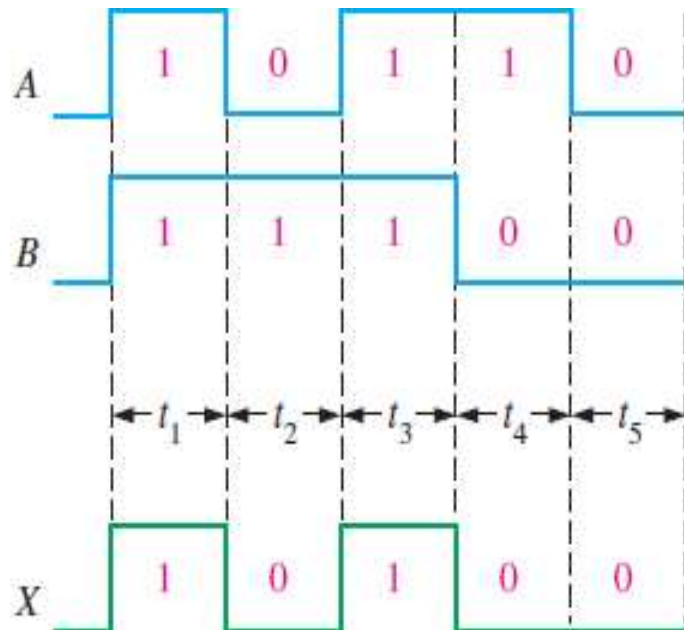
- An **AND gate** produces a **HIGH** output *only* when *all* of the inputs are **HIGH**. When any of the inputs is **LOW**, the output is **LOW**.



Inputs		Output
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

1 = HIGH, 0 = LOW

- **Timing Diagram:**



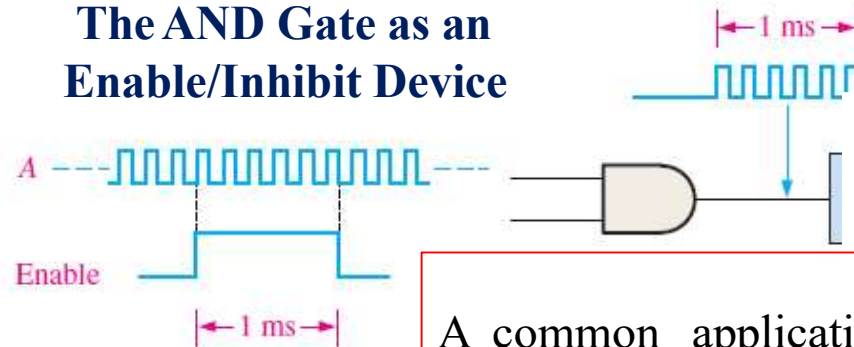
$$\begin{aligned}
 0 \cdot 0 &= 0 \\
 0 \cdot 1 &= 0 \\
 1 \cdot 0 &= 0 \\
 1 \cdot 1 &= 1
 \end{aligned}$$

The logical AND function of two variables (A, B) is represented mathematically either by placing a dot between the two variables, as $A \cdot B$, or by simply writing the adjacent letters without the dot, as AB .

$$X = A \cdot B = AB$$

Application

The AND Gate as an Enable/Inhibit Device



A common application of the AND gate is to **enable** (that is, to allow) the passage of a signal (pulse waveform) from one point to another at certain times and to **inhibit** (prevent) the passage at other times.

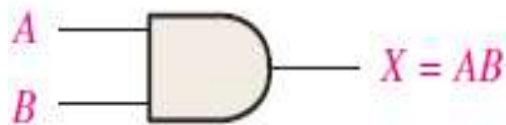
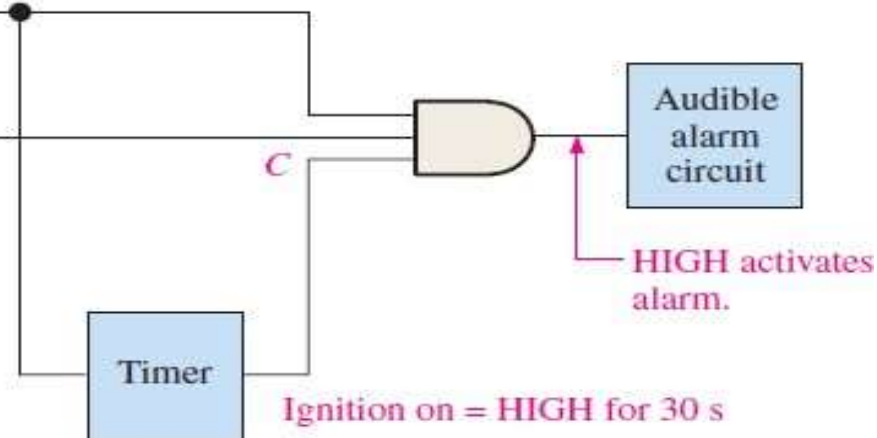
HIGH = On
LOW = Off

HIGH = Unbuckled
LOW = Buckled

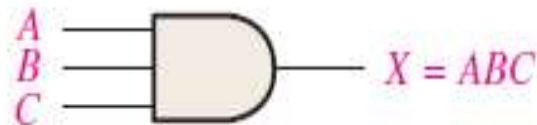
Ignition switch A

Seat belt B

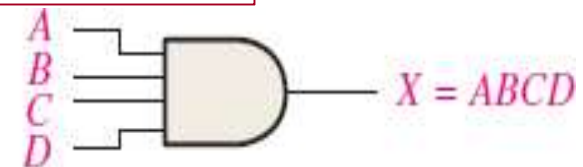
A Seat Belt Alarm System



(a)



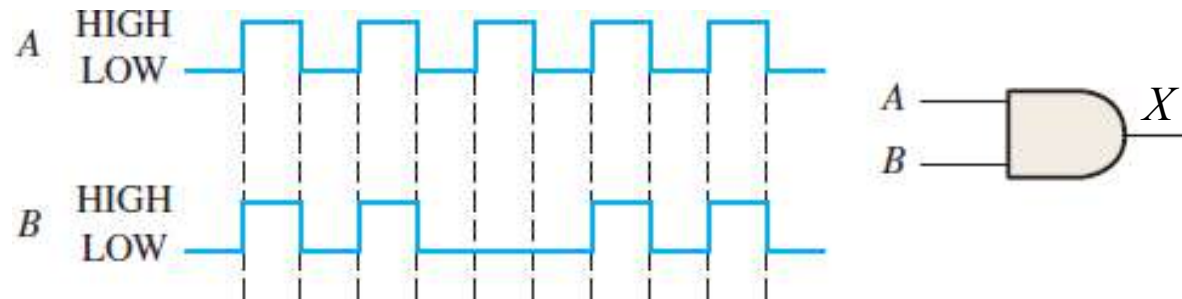
(b)



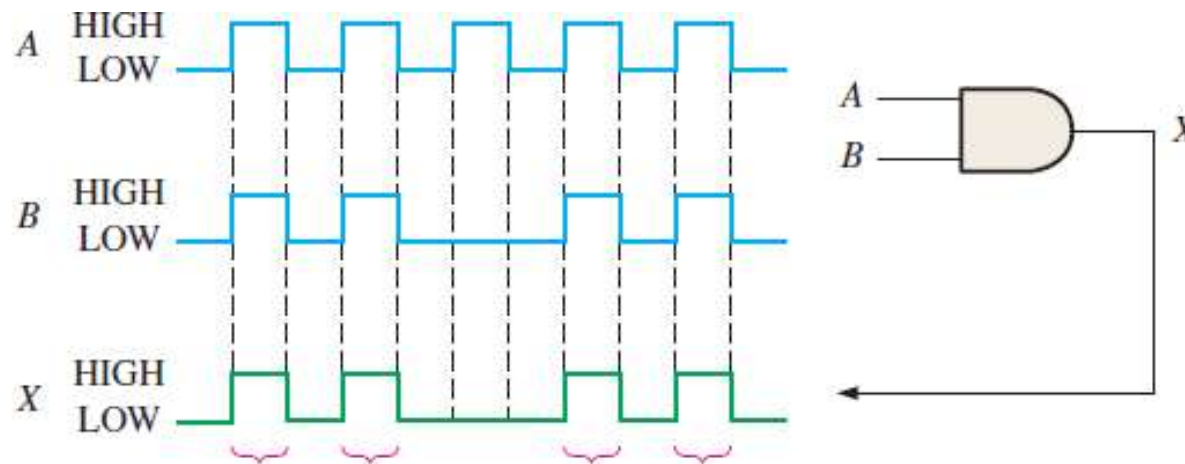
(c)

Example:

1. If two waveforms, A and B , are applied to the AND gate inputs as shown below, what is the resulting output waveform?



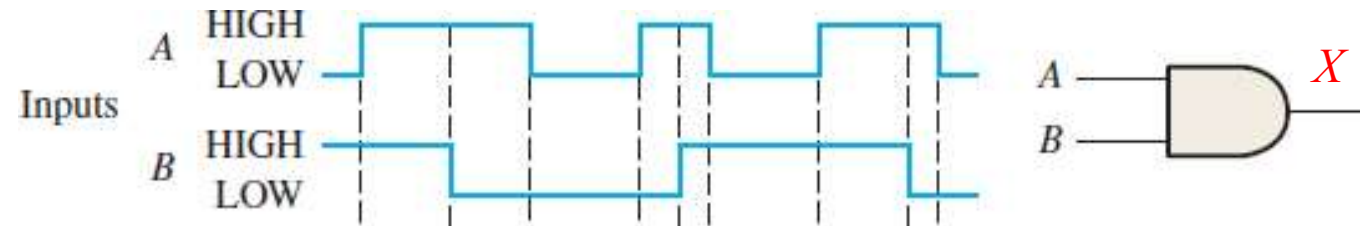
Solution



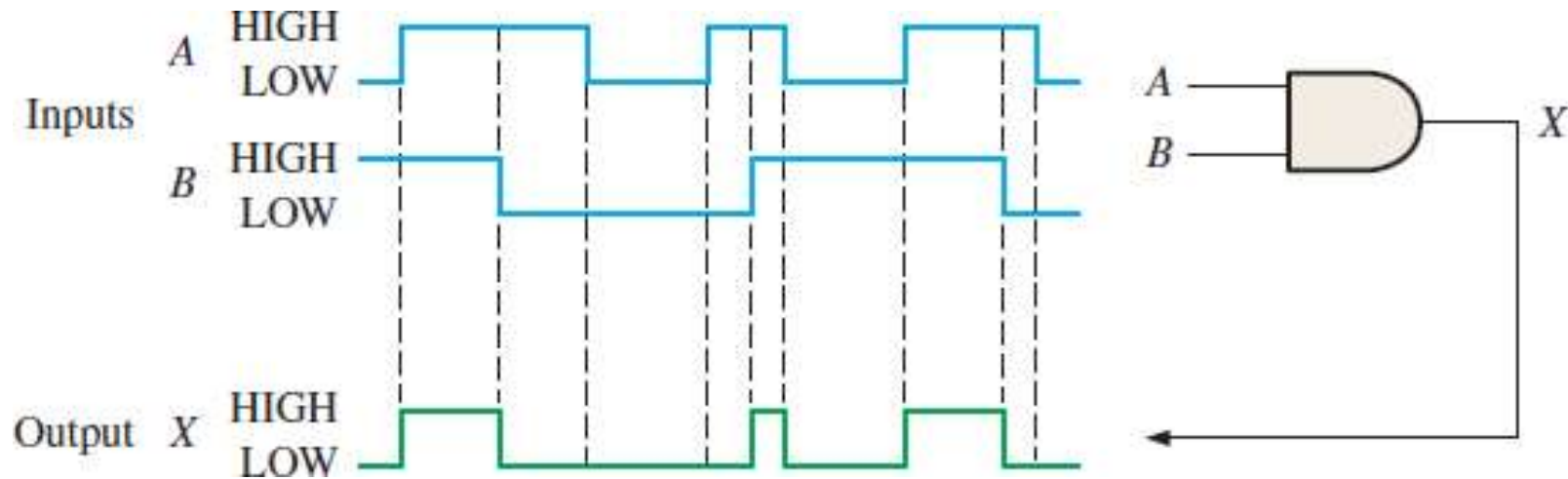
A and B are both HIGH during these four time intervals; therefore, X is HIGH.

The output waveform X is HIGH only when both A and B waveforms are HIGH

2. For the two input waveforms, A and B , in Figure below, show the output waveform with its proper relation to the inputs.

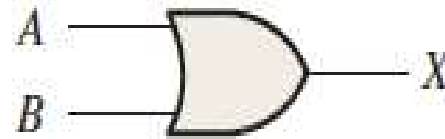


Solution



OR Gate

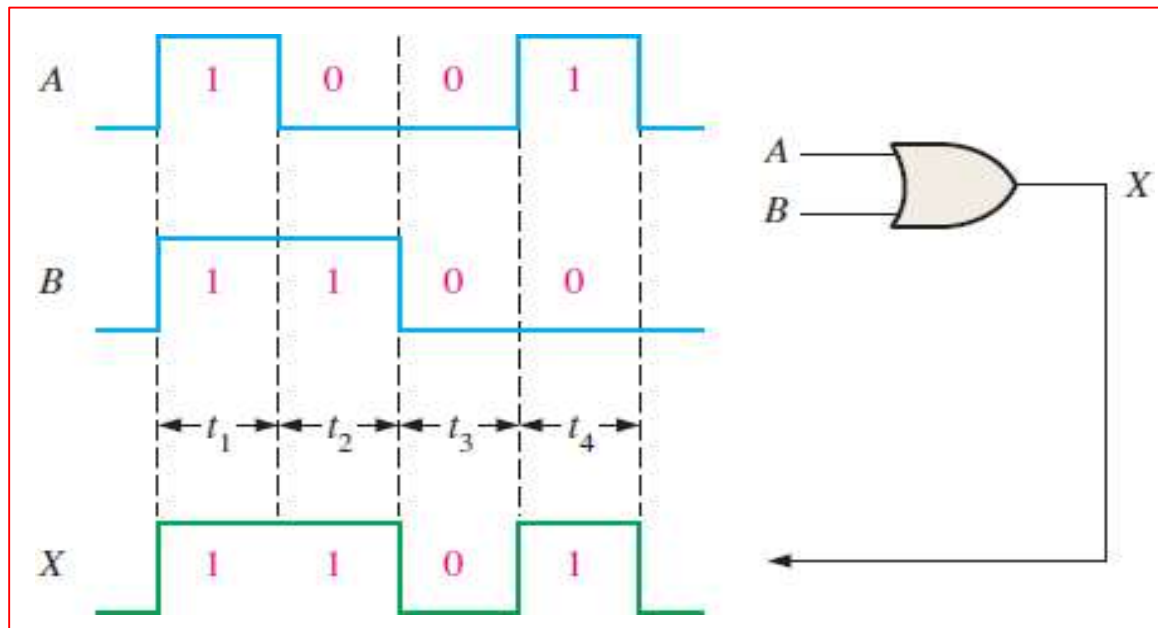
- An **OR gate** produces a **HIGH** on the output when any of the inputs is **HIGH**. The output is **LOW** only when all of the inputs are **LOW**.



Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

1 = HIGH, 0 = LOW

- **Timing Diagram:**



$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 1$$

The logical OR function of two variables is represented mathematically by a + between the two variables, for example, $A + B$. The plus sign is read as “OR.”

$$X = A + B$$

Application

Intrusion detection and alarm system

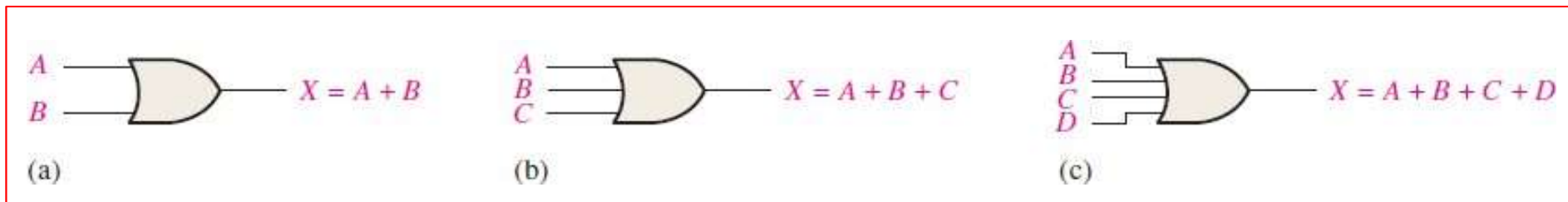
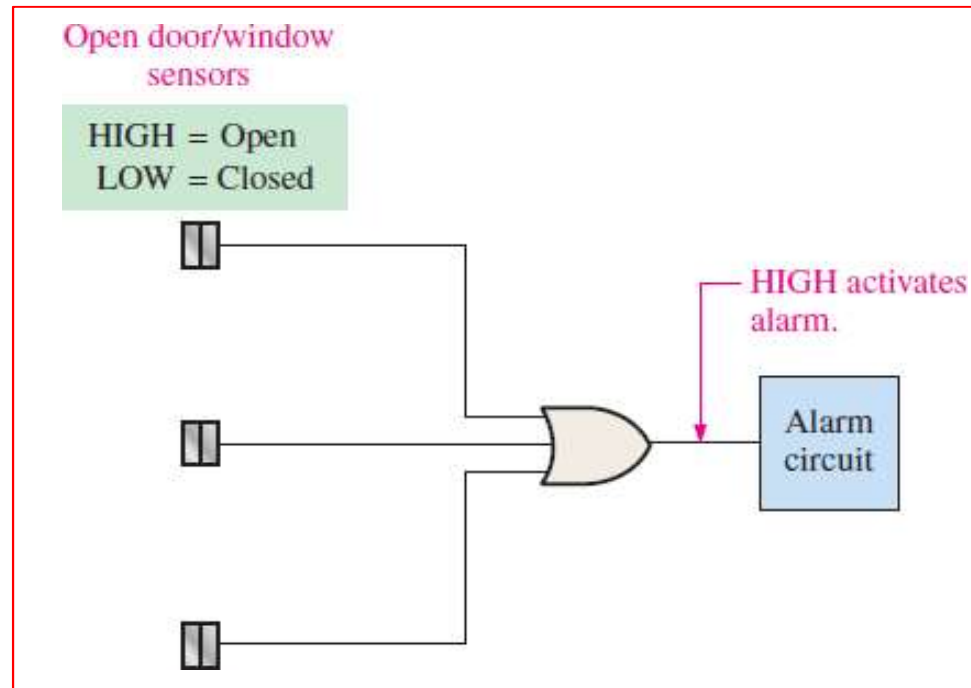
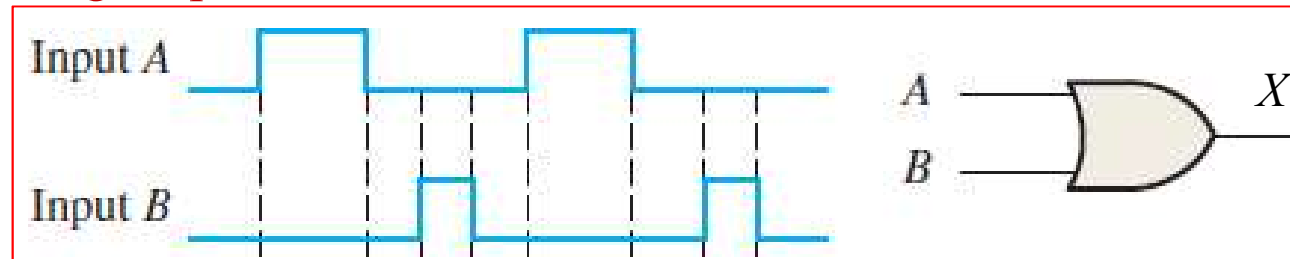


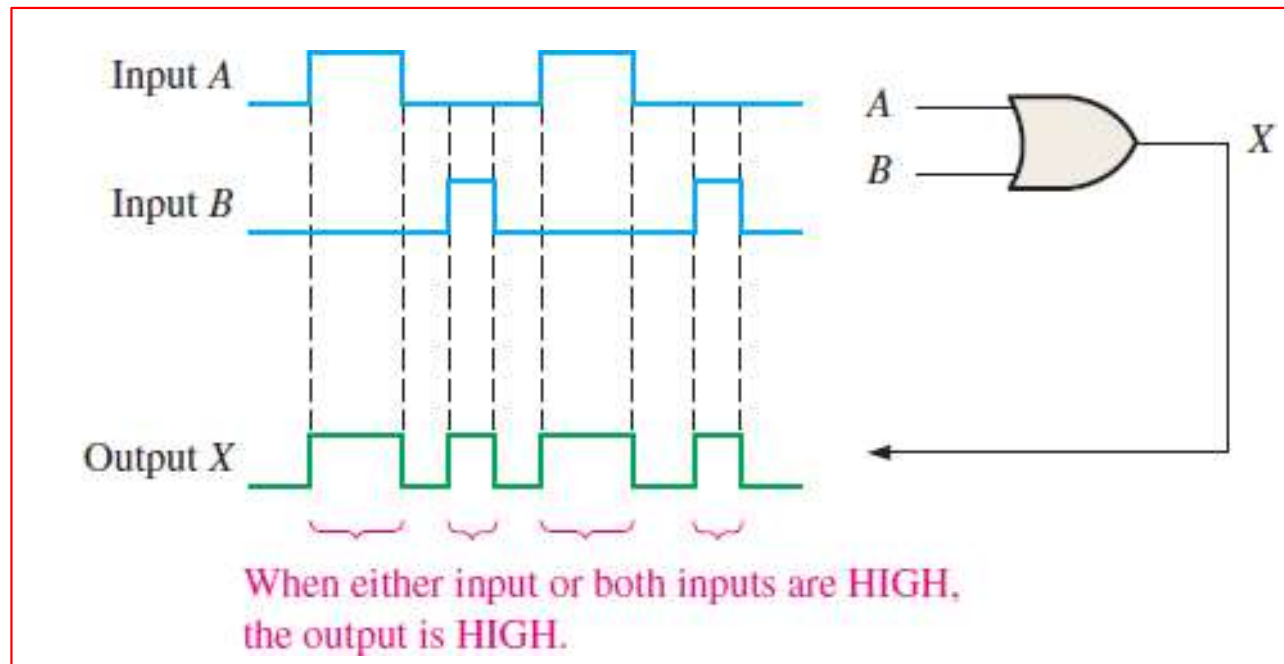
Fig: Boolean expressions for OR gates with two, three, and four inputs.

Example:

1. If the two input waveforms, A and B , in Figure below are applied to the OR gate, what is the resulting output waveform?

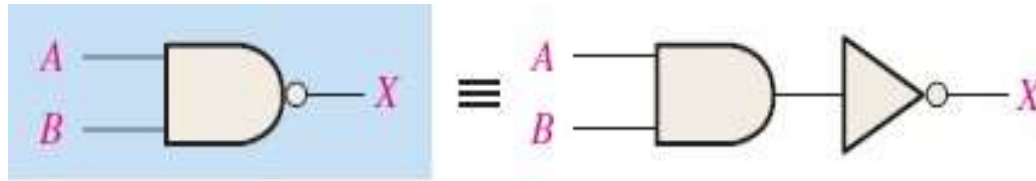


Solution:



NAND Gate

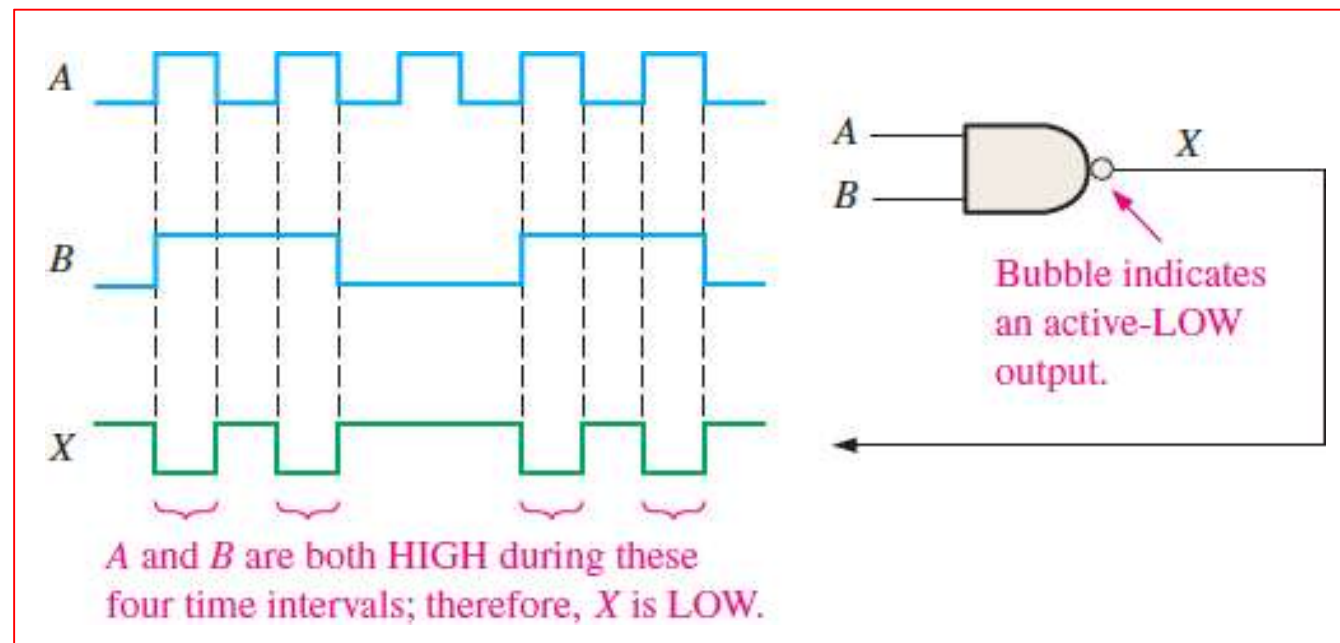
- A **NAND gate** produces a **LOW** output only when all the inputs are **HIGH**. When any of the inputs is **LOW**, the output will be **HIGH**.



Inputs		Output
A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

1 = HIGH, 0 = LOW.

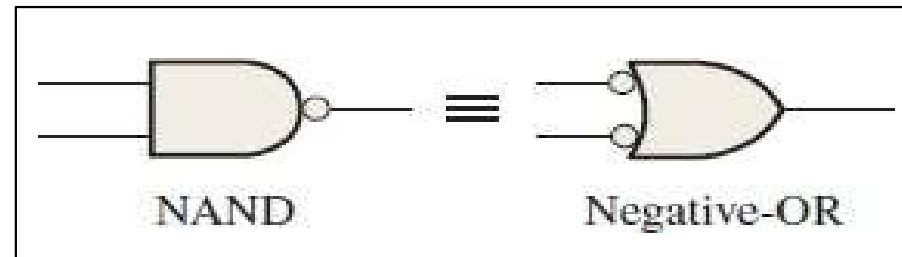
- **Timing Diagram:**



- The Boolean expression for the output of a 2-input NAND gate is:

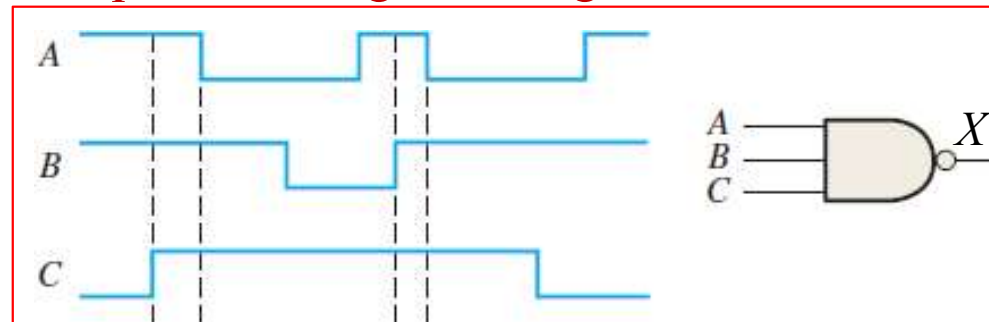
$$X = \overline{A \cdot B} = \overline{AB}$$

➤ Negative-OR Equivalent Operation of a NAND Gate:

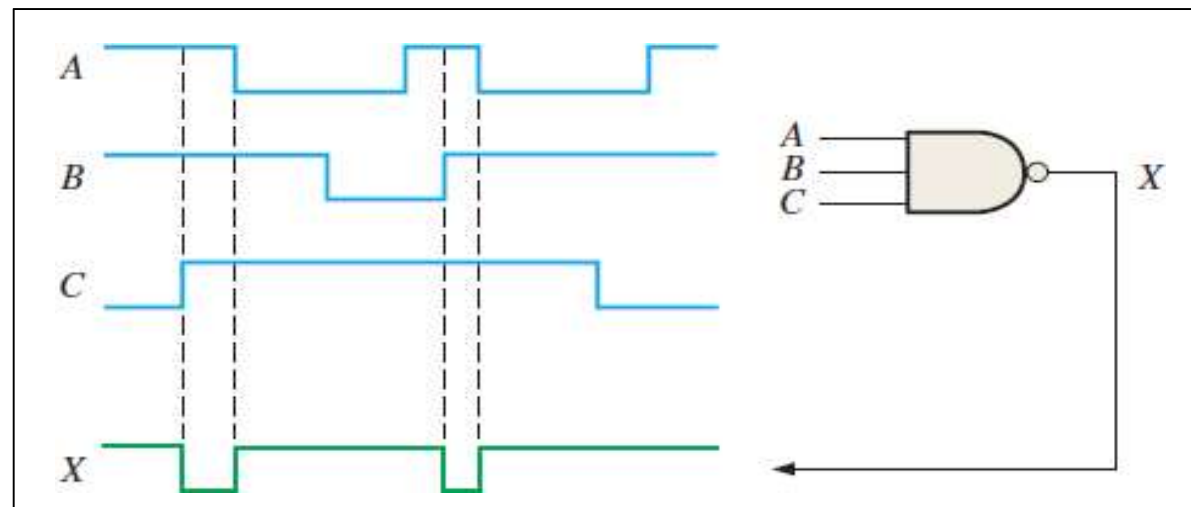


Example:

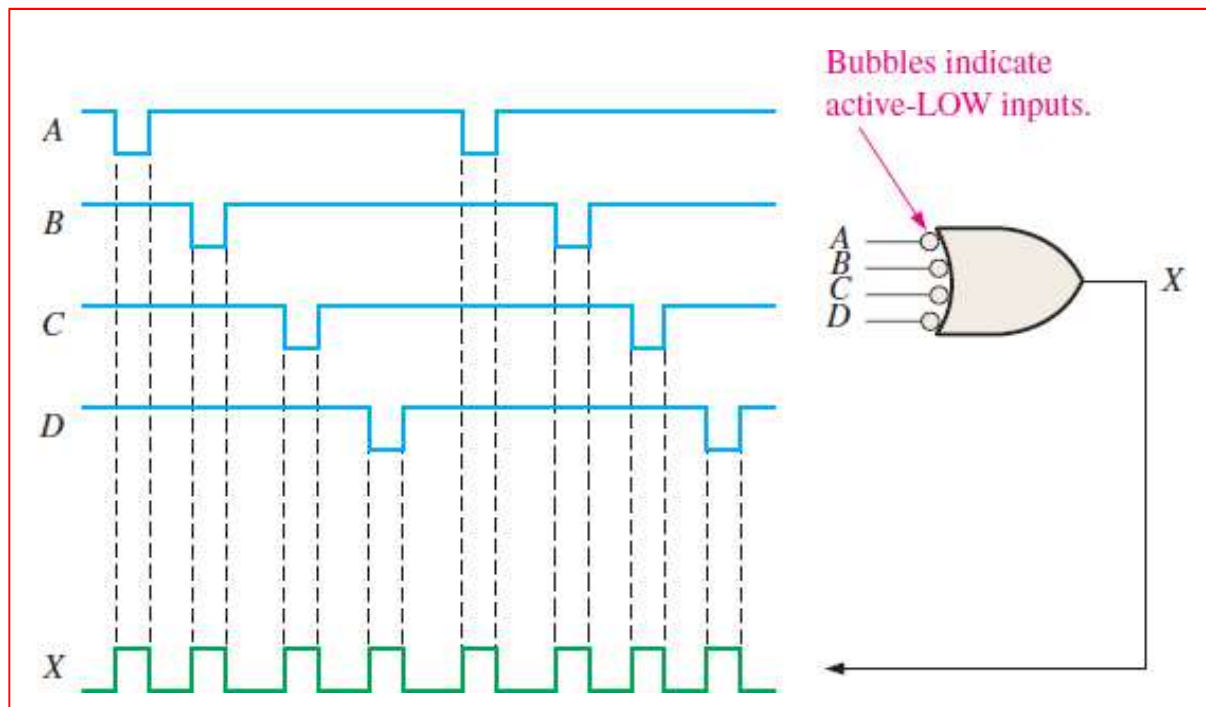
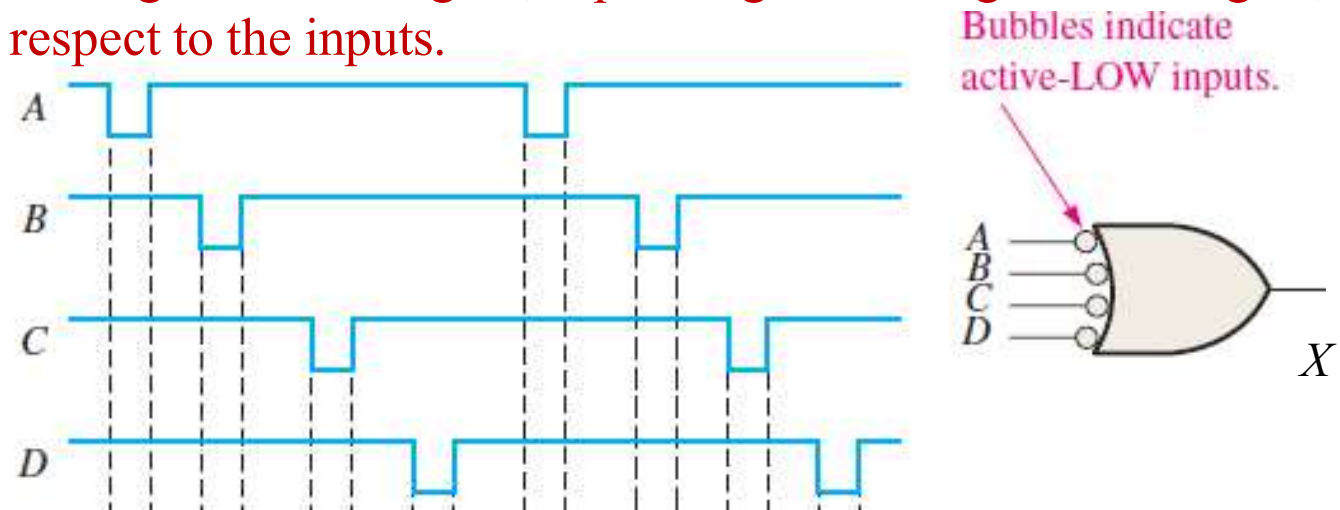
Show the output waveform for the 3-input NAND gate in Figure below with its proper time relationship to the inputs.



Solution:



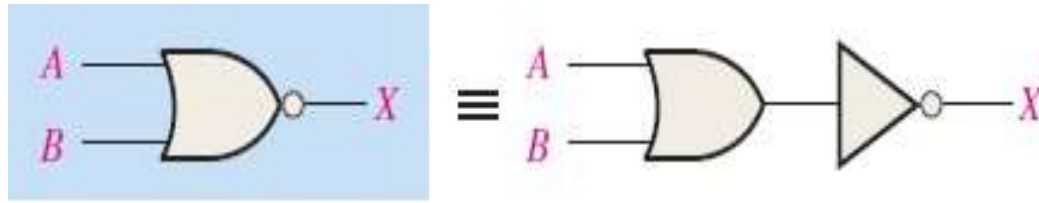
For the 4-input NAND gate in Figure, operating as a negative-OR gate, determine the output with respect to the inputs.



Solution:

NOR Gate

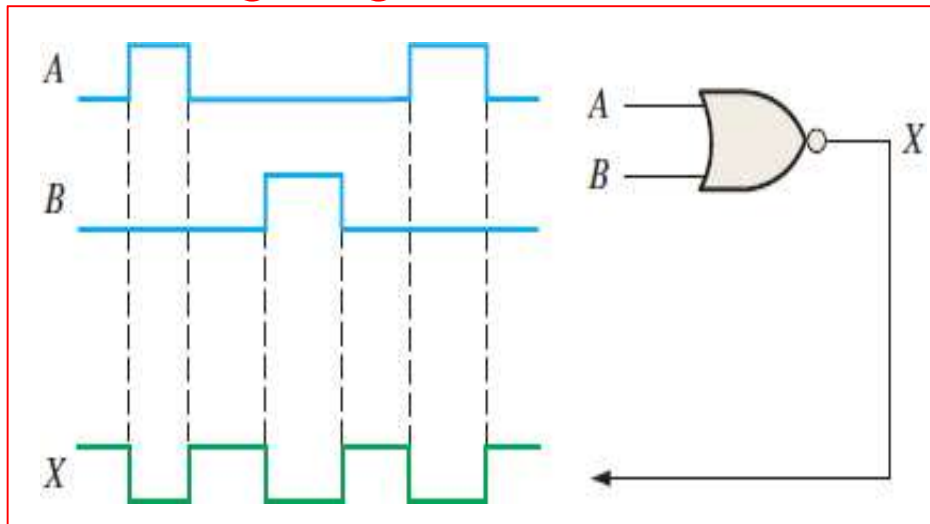
- A **NOR gate** produces a **LOW** output when any of its inputs is **HIGH**. Only when all of its inputs are **LOW** is the output **HIGH**.



Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

1 = HIGH, 0 = LOW.

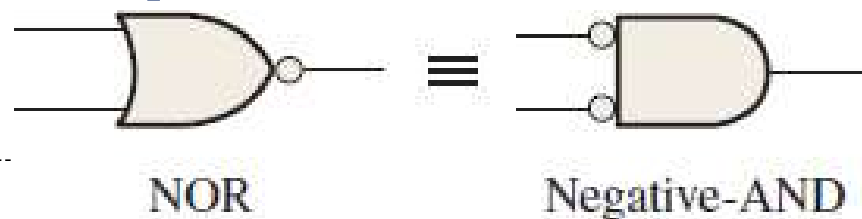
- **Timing Diagram:**



The Boolean expression for the output of a 2-input NOR gate is:

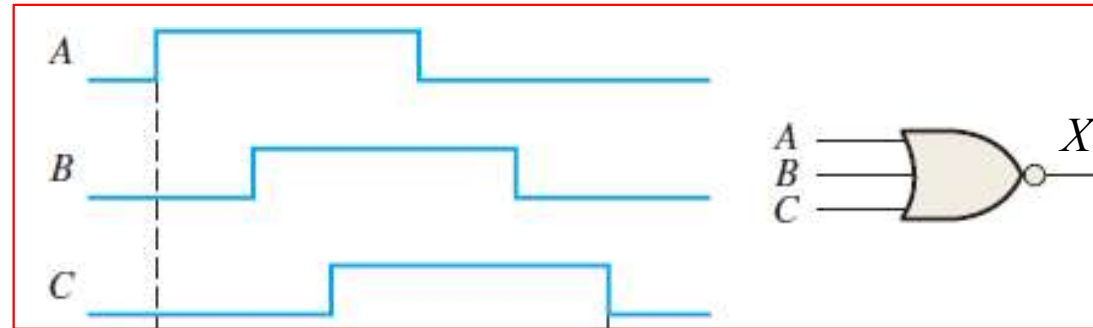
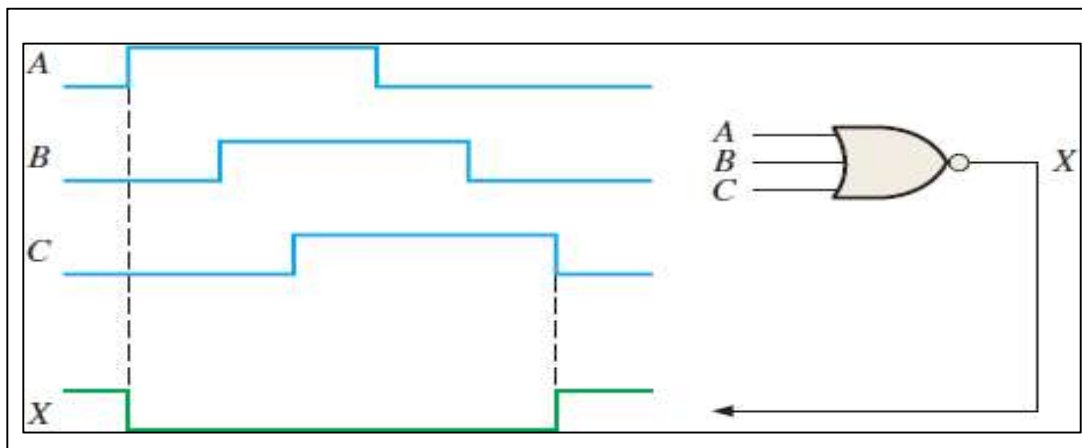
$$X = \overline{A + B}$$

- **Negative-AND Equivalent Operation of the NOR Gate:**



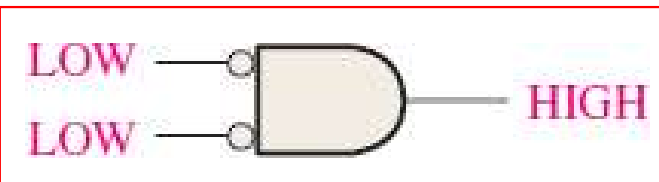
Example:

Show the output waveform for the 3-input NOR gate in Figure with the proper time relation to the inputs.

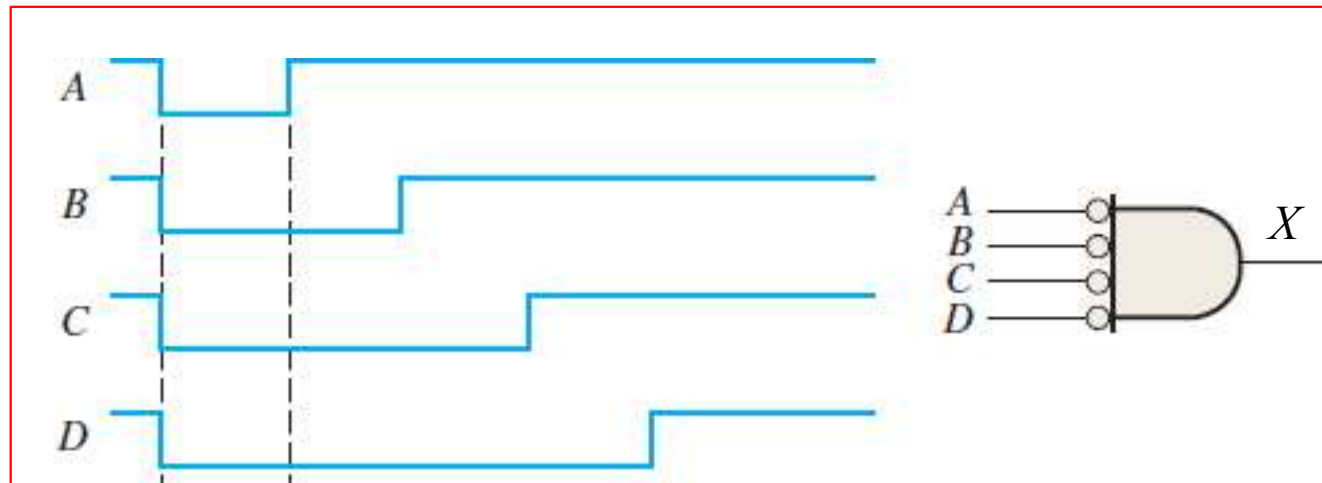
**Solution:**

A device is needed to indicate when two LOW levels occur simultaneously on its inputs and to produce a HIGH output as an indication. Specify the device.

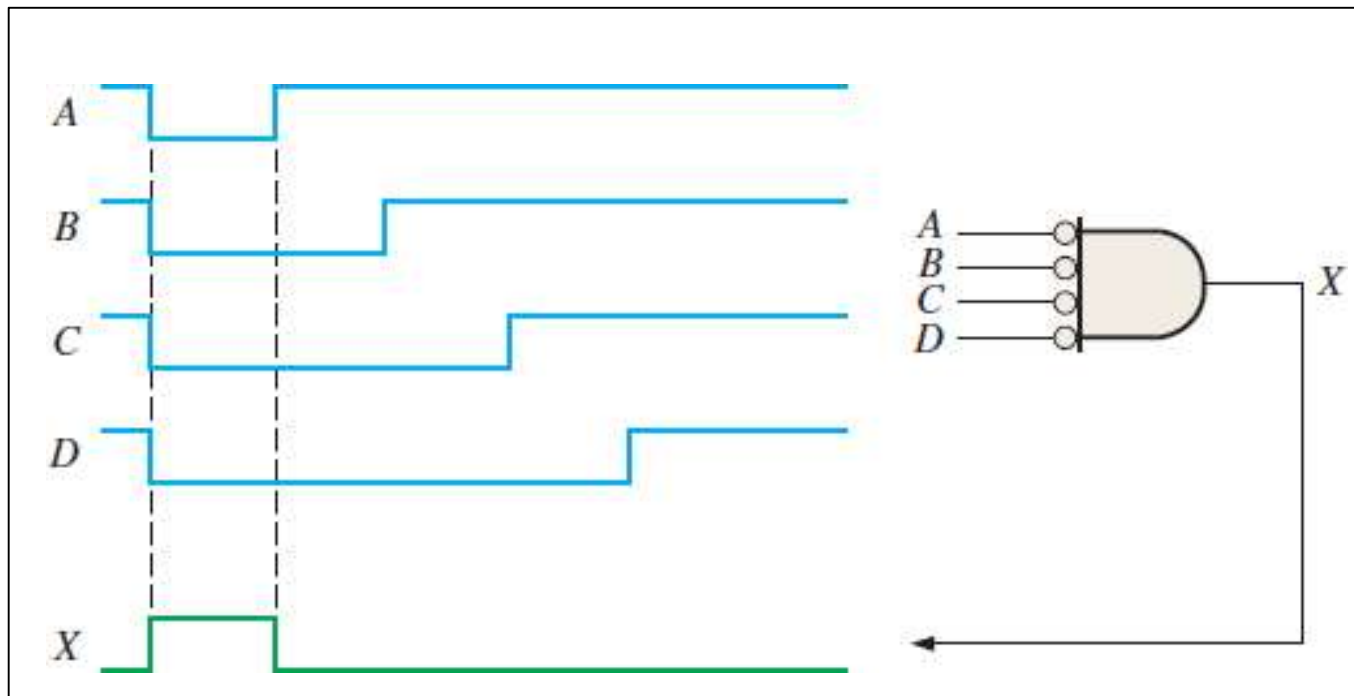
Solution: A 2-input NOR gate operating as a negative-AND gate is required to produce a HIGH output when both inputs are LOW, as shown below:



For the 4-input NOR gate operating as a negative-AND in Figure, determine the output relative to the inputs.



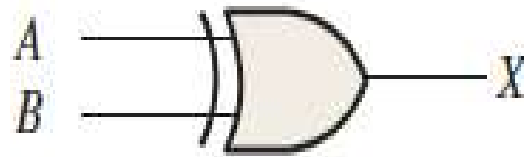
Solution



Exclusive-OR Gate

- For an **exclusive-OR** gate, output X is HIGH when input A is LOW and input B is HIGH, or when input A is HIGH and input B is LOW; X is LOW when A and B are both HIGH or both LOW.

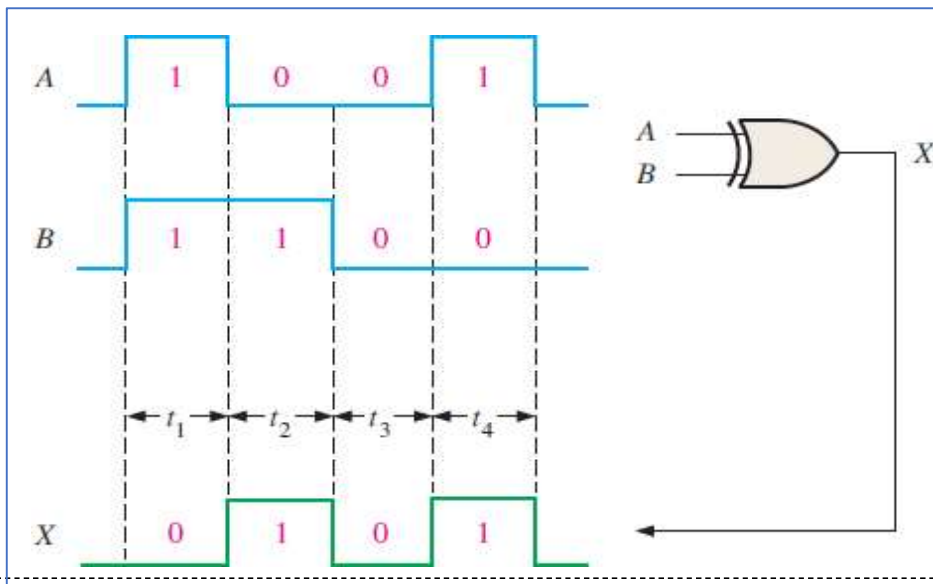
Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	0



The Boolean expression for the output of a 2-input XOR gate is:

$$X = A \oplus B = \bar{A}B + A\bar{B}$$

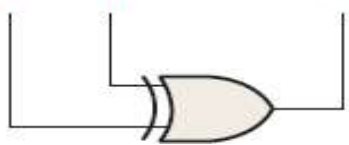
- **Timing Diagram:**



Application: An exclusive-OR gate can be used as a two-bit modulo-2 adder.

An XOR gate used to add two bits.

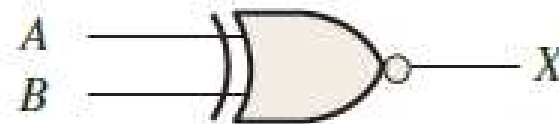
Input Bits		Output (Sum)
A	B	Σ
0	0	0
0	1	1
1	0	1
1	1	0 (without the 1 carry bit)



Exclusive-NOR Gate

- For an **exclusive-NOR** gate, output X is LOW when input A is LOW and input B is HIGH, or when A is HIGH and B is LOW; X is HIGH when A and B are both HIGH or both LOW.

Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

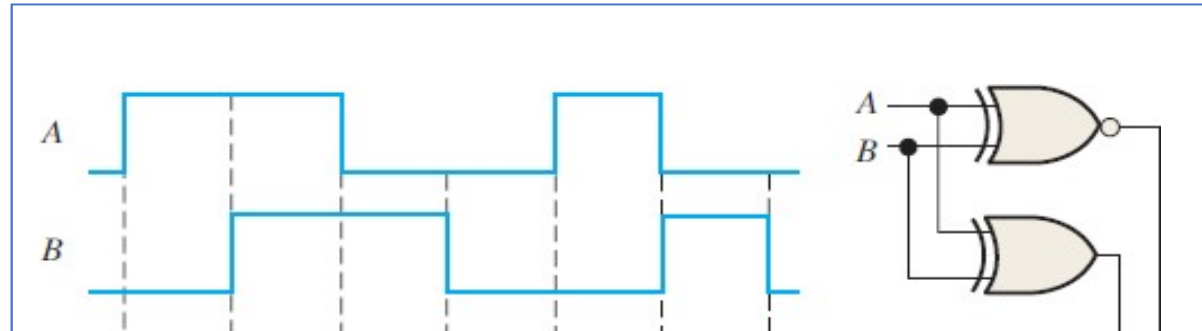


The Boolean expression for the output of a 2-input XNOR gate is:

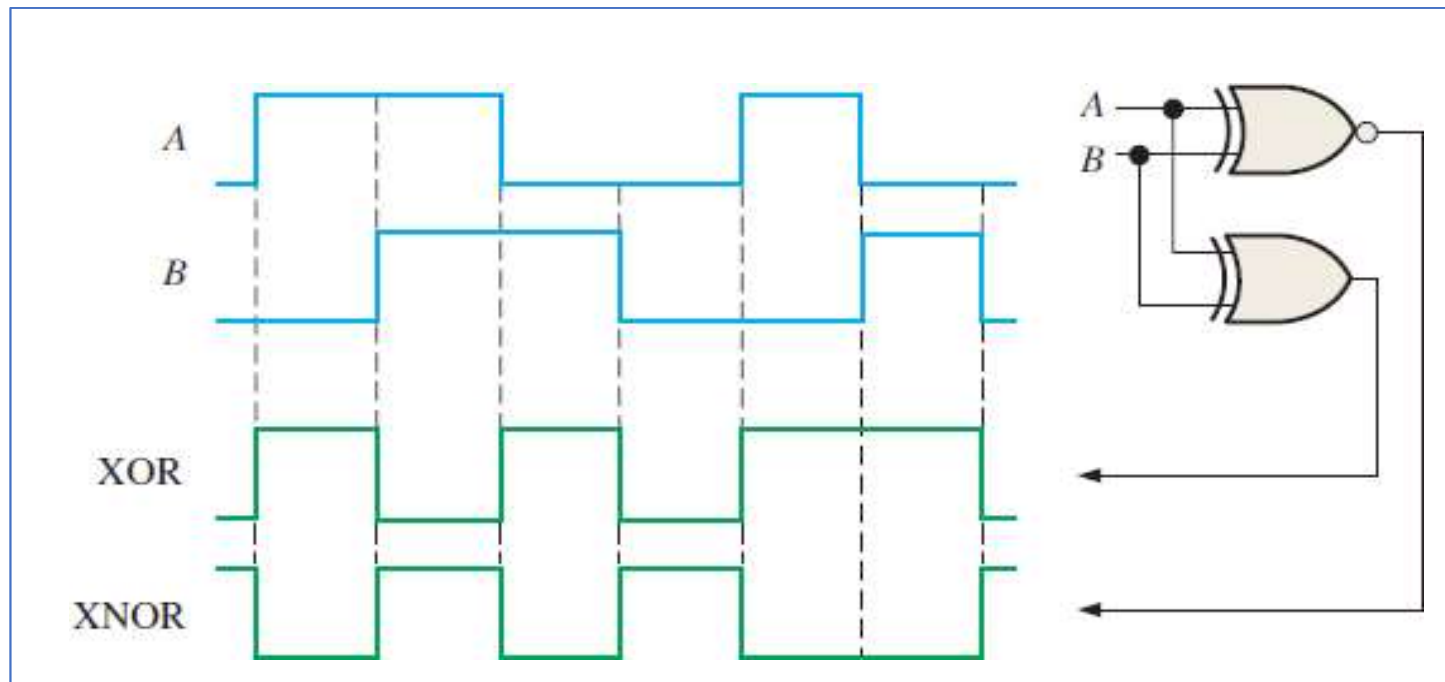
$$X = A \odot B = AB + \bar{A}\bar{B}$$

Example:

Determine the output waveforms for the XOR gate and for the XNOR gate, for the given the input waveforms, A and B as shown in Figure below.

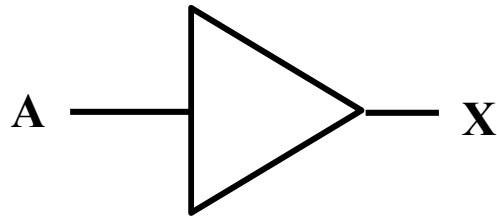


Solution



Buffer Gate

- For a **buffer gate**, **output X is LOW** when **input A is LOW** and **output X is HIGH**, or when **A is HIGH**.
- Buffer circuits are symbolized by a triangle symbol with no inverter “bubble”.



The Boolean expression for the output of a Buffer gate is:

$$X = A$$

Input	Output
A	X
0	0
1	1

Boolean Algebra

- Boolean algebra is the mathematics of digital logic.
- Like any other deductive mathematical system, **Boolean algebra may be defined** with a **set of elements**, a **set of operators**, and a **number of unproved axioms or postulates**.
- A **variable** is a symbol (usually an italic uppercase letter or word) used to represent an action, a condition, or data.
- **Complement:** The **complement** is the inverse of a variable and is indicated by a bar over the variable (overbar). Example: The complement of the variable A is $\bar{A}$. If $A = 1$, then $\bar{A} = 0$. If $A = 0$, then $\bar{A} = 1$.
 - Sometimes a prime symbol rather than an overbar is used to denote the complement of a variable; for example, B' indicates the complement of B .
- **Boolean addition** is equivalent to the OR operation.
- **Boolean multiplication** is equivalent to the AND operation.

- **Postulates** of a mathematical system form the basic assumptions from which it is possible to deduce the rules, theorems, and properties of the system.
- The most common postulates used to formulate various algebraic structures are
 - Closure
 - Associative law
 - Commutative law
 - Identity element
 - Inverse
 - Distributive law

- **Closure:** A set S is closed with respect to a binary operator if, for every pair of elements of S , the binary operator specifies a rule for obtaining a unique element of S .
 - For example, the set of natural numbers $N = \{1, 2, 3, 4, \dots\}$ is closed with respect to the binary operator plus (+) by the rules of arithmetic addition, since for any $a, b \in N$ we obtain a unique $c \in N$ by the operation $a + b = c$.
- **Associative law:** A binary operator $*$ on a set S is said to be associative whenever:
 $(x * y) * z = x * (y * z)$ for all $x, y, z \in S$
- **Commutative law:** A binary operator $*$ on a set S is said to be commutative whenever: $x * y = y * x$ for all $x, y \in S$
- **Identity element:** A set S is said to have an identity element with respect to a binary operation $*$ on S if there exists an element $e \in S$ with the property:
 $e * x = x * e = x$ for every $x \in S$
- **Distributive law:** If $*$ and $\cdot$ are two binary operators on a set S , $*$ is said to be distributive over $\cdot$ whenever: $x * (y \cdot z) = (x * y) \cdot (x * z)$

Axiomatic definition of Boolean Algebra

- In 1854, George Boole developed an algebraic system now called *Boolean algebra*.
- In 1938, Claude E. Shannon introduced a two-valued Boolean algebra called *switching algebra*
- *Switching algebra* that represented the properties of bistable electrical switching circuits.
- For the formal definition of Boolean algebra, we shall employ the **postulates** formulated by E. V. Huntington in 1904.
- Boolean algebra is an algebraic structure defined by a set of elements, B , together with two binary operators, $+$ and $\cdot$, provided that a set of postulates are satisfied.

Postulates

1. (a) The structure is closed with respect to the operator $+$.
(b) The structure is closed with respect to the operator $\cdot$.
2. (a) The element 0 is an identity element with respect to $+$: $x + 0 = 0 + x = x$.
(b) The element 1 is an identity element with respect to $\cdot$: $x \cdot 1 = 1 \cdot x = x$.
3. (a) The structure is commutative with respect to $+$: $x + y = y + x$.
(b) The structure is commutative with respect to $\cdot$: $x \cdot y = y \cdot x$.
4. (a) The operator $\cdot$ is distributive over $+$: $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$.
(b) The operator $+$ is distributive over $\cdot$: $x + (y \cdot z) = (x + y) \cdot (x + z)$.
5. For every element $x \in B$, there exists an element $x' \in B$ (called the complement of x) such that (a) $x + x' = 1$ and (b) $x \cdot x' = 0$.
6. There exist at least two elements $x, y \in B$ such that $x \neq y$.

Two-Valued Boolean Algebra

- A two-valued Boolean algebra is defined on a set of two elements, $B = \{0, 1\}$, with rules for the two binary operators $+$ and $\cdot$ as shown below

x	y	$x \cdot y$
0	0	0
0	1	0
1	0	0
1	1	1

x	y	$x + y$
0	0	0
0	1	1
1	0	1
1	1	1

x	x'
0	1
1	0

❖ Duality:

Basic Theorems and Properties of Boolean Algebra

- ▣ The principle of *duality* is an important concept. This says that if an expression is valid in Boolean algebra, the dual of that expression is also valid.
- ▣ To form the dual of an expression, replace all (+) operators with (·) operators, all (·) operators with (+) operators, all ones with zeros, and all zeros with ones.
- ▣ Following the replacement rules...
$$a(b + c) = ab + ac$$
- ▣ Form the dual of the expression
$$a + (bc) = (a + b)(a + c)$$
- ▣ Take care not to alter the location of the parentheses if they are present.

Postulates and Theorems of Boolean Algebra

Postulate 2	(a) $x + 0 = x$	(b) $x \cdot 1 = x$
Postulate 5	(a) $x + x' = 1$	(b) $x \cdot x' = 0$
Theorem 1	(a) $x + x = x$	(b) $x \cdot x = x$
Theorem 2	(a) $x + 1 = 1$	(b) $x \cdot 0 = 0$
Theorem 3, involution	$(x')' = x$	
Postulate 3, commutative	(a) $x + y = y + x$	(b) $xy = yx$
Theorem 4, associative	(a) $x + (y + z) = (x + y) + z$	(b) $x(yz) = (xy)z$
Postulate 4, distributive	(a) $x(y + z) = xy + xz$	(b) $x + yz = (x + y)(x + z)$
Theorem 5, DeMorgan	(a) $(x + y)' = x'y'$	(b) $(xy)' = x' + y'$
Theorem 6, absorption	(a) $x + xy = x$	(b) $x(x + y) = x$

THEOREM 1(a): $x + x = x$.

Statement	Justification
$x + x = (x + x) \cdot 1$	postulate 2(b)
$= (x + x)(x + x')$	5(a)
$= x + xx'$	4(b)
$= x + 0$	5(b)
$= x$	2(a)

THEOREM 1(b): $x \cdot x = x$.

Statement	Justification
$x \cdot x = xx + 0$	postulate 2(a)
$= xx + xx'$	5(b)
$= x(x + x')$	4(a)
$= x \cdot 1$	5(a)
$= x$	2(b)

THEOREM 2(a): $x + 1 = 1$.

Statement	Justification
$x + 1 = 1 \cdot (x + 1)$	postulate 2(b)
$= (x + x')(x + 1)$	5(a)
$= x + x' \cdot 1$	4(b)
$= x + x'$	2(b)
$= 1$	5(a)

THEOREM 2(b): $x \cdot 0 = 0$ by duality.

THEOREM 3: $(x')' = x$. From postulate 5, we have $x + x' = 1$ and $x \cdot x' = 0$, which together define the complement of x . The complement of x' is x and is also $(x')'$.

THEOREM 6(a): $x + xy = x$.

Statement	Justification
$x + xy = x \cdot 1 + xy$	postulate 2(b)
$= x(1 + y)$	4(a)
$= x(y + 1)$	3(a)
$= x \cdot 1$	2(a)
$= x$	2(b)

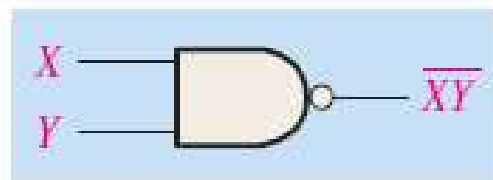
THEOREM 6(b): $x(x + y) = x$ by duality.

DeMorgan's Theorems

➤ *DeMorgan's first theorem is stated as follows:*

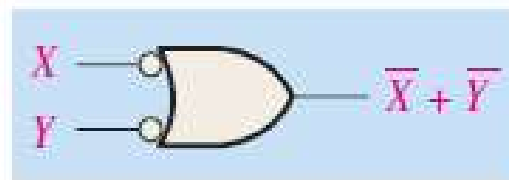
- The complement of a product of variables is equal to the sum of the complements of the variables.
- Stated another way, The complement of two or more ANDed variables is equivalent to the OR of the complements of the individual variables.
- The formula for expressing this theorem for two variables is: $\overline{XY} = \overline{X} + \overline{Y}$

Inputs		Output	
X	Y	$\overline{XY}$	$\overline{X} + \overline{Y}$
0	0	1	1
0	1	1	1
1	0	1	1
1	1	0	0



NAND

≡



Negative-OR

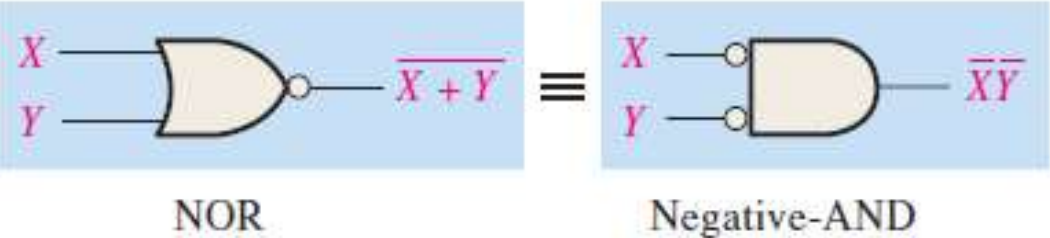
DeMorgan's Theorems

➤ *DeMorgan's second theorem is stated as follows:*

- The complement of a sum of variables is equal to the product of the complements of the variables.
- Stated another way, The complement of two or more ORed variables is equivalent to the AND of the complements of the individual variables.
- The formula for expressing this theorem for two variables is:

$$\overline{X + Y} = \overline{X} \overline{Y}$$

Inputs		Output	
X	Y	$\overline{X + Y}$	$\overline{X} \overline{Y}$
0	0	1	1
0	1	0	0
1	0	0	0
1	1	0	0



1. Apply DeMorgan's theorems to the expressions $\overline{XYZ}$ and $\overline{X + Y + Z}$.

Solution

$$\begin{aligned}\overline{XYZ} &= \overline{X} + \overline{Y} + \overline{Z} \\ \overline{X + Y + Z} &= \overline{X} \overline{Y} \overline{Z}\end{aligned}$$

2. Apply DeMorgan's theorems to the expressions $\overline{WXYZ}$ and $\overline{W + X + Y + Z}$.

Solution

$$\begin{aligned}\overline{WXYZ} &= \overline{W} + \overline{X} + \overline{Y} + \overline{Z} \\ \overline{W + X + Y + Z} &= \overline{W} \overline{X} \overline{Y} \overline{Z}\end{aligned}$$

3. Apply DeMorgan's theorems to each of the following expressions:

(a) $\overline{(A + B + C)D}$

(b) $\overline{ABC + DEF}$

(c) $\overline{A\overline{B} + \overline{C}D + EF}$

Solution

- (a) Let $A + B + C = X$ and $D = Y$. The expression $\overline{(A + B + C)D}$ is of the form $\overline{XY} = \overline{X} + \overline{Y}$ and can be rewritten as

$$\overline{(A + B + C)D} = \overline{A + B + C} + \overline{D}$$

Next, apply DeMorgan's theorem to the term $\overline{A + B + C}$.

$$\overline{A + B + C} + \overline{D} = \overline{A} \overline{B} \overline{C} + \overline{D}$$

- (b) Let $ABC = X$ and $DEF = Y$. The expression $\overline{ABC + DEF}$ is of the form $\overline{X + Y} = \overline{X} \overline{Y}$ and can be rewritten as

$$\overline{ABC + DEF} = (\overline{ABC})(\overline{DEF})$$

Next, apply DeMorgan's theorem to each of the terms $\overline{ABC}$ and $\overline{DEF}$.

$$(\overline{ABC})(\overline{DEF}) = (\overline{A} + \overline{B} + \overline{C})(\overline{D} + \overline{E} + \overline{F})$$

- (c) Let $\overline{AB} = X$, $\overline{CD} = Y$, and $EF = Z$. The expression $\overline{\overline{AB} + \overline{CD} + EF}$ is of the form $\overline{X + Y + Z} = \overline{X} \overline{Y} \overline{Z}$ and can be rewritten as

$$\overline{\overline{AB} + \overline{CD} + EF} = (\overline{\overline{AB}})(\overline{\overline{CD}})(\overline{EF})$$

Next, apply DeMorgan's theorem to each of the terms $\overline{\overline{AB}}$, $\overline{\overline{CD}}$, and $\overline{EF}$.

$$(\overline{\overline{AB}})(\overline{\overline{CD}})(\overline{EF}) = (\overline{A} + B)(C + \overline{D})(\overline{E} + \overline{F})$$

EXAMPLE: Using basic Boolean theorem prove:

(a) $(x + y)(x + z) = x + yz$

Soln: $(x + y)(x + z)$

$$= x \cdot x + x \cdot z + y \cdot x + y \cdot z$$

since $x \cdot x = x$

$$= x + xz + yx + yz$$

$$= x(1 + z) + yx + yz$$

since $(1 + z) = 1$

$$= x + yx + yz$$

$$= x(1 + y) + yz$$

since $(1 + y) = 1$

$$= x + yz \text{ (Proved)}$$

(b) $xy + xz + yz' = xz + yz'$

Soln: $xy + xz + yz'$

$$= xy(z + z') + xz(y + y') + yz'(x + x')$$

since $x + x' = 1$

$$= xyz + xyz' + xyz + xy'z + xyz' + x'yz'$$

$$= xyz + xyz' + xy'z + x'yz'$$

since $xyz + xyz = xyz$

$$= xyz + xy'z + xyz' + x'yz'$$

rearranging

$$= xz(y + y') + yz'(x + x')$$

since $y + y' = 1$ and $x + x' = 1$

$$= xz + yz' \text{ (Proved)}$$

EXAMPLE : Simplify the following Boolean functions to a minimum number of literals.

1. $x + x' y = (x + x')(x + y) = 1 \cdot (x + y) = x + y$

2. $x (x' + y) = xx' + xy = 0 + xy = xy$

3. $x' y' z + x' yz + xy' = x' z(y' + y) + xy' = x' z + xy'$

4. $xy + x' z + yz = xy + x' z + yz(x + x')$
 $= xy + x' z + xyz + x' yz$
 $= xy(1 + z) + x' z(1 + y)$
 $= xy + x' z$

5. $(x + y)(x' + z)(y + z) = (x + y)(x' + z)$ by duality from function 4.

Complement of a function

$$\begin{aligned}(A + B + C)' &= (A + X)' \\ &= A' X' \\ &= A' \cdot (B + C)' \\ &= A' \cdot (B' C') \\ &= A' B' C'\end{aligned}$$

$$\text{let } B + C = X$$

by theorem 5(a) (De Morgan)

substitute $B + C = X$

by theorem 5(a) (De Morgan)

by theorem 4(b) (associative)

$$(A + B + C + D + \dots + F)' = A' B' C' D' \dots F'$$

$$(ABCD \dots F)' = A' + B' + C' + D' + \dots + F'$$

➤ The generalized form of De Morgan's theorem states that the complement of a function is obtained by interchanging AND and OR operators and complementing each literal.

EXAMPLE : Find the complement of the functions $F_1 = x' yz' + x' y' z$ and $F_2 = x (y'z' + yz)$. Applying De Morgan's theorem as many times as necessary, the complements are obtained as follows:

$$F'_1 = (x' yz' + x' y' z)' = (x' yz')'(x' y' z)' = (x + y' + z)(x + y + z')$$

$$\begin{aligned} F'_2 &= [x (y' z' + yz)]' = x' + (y' z' + yz)' = x' + (y' z')' \cdot (yz)' \\ &= x' + (y + z)(y' + z') \end{aligned}$$